

3

The psychological reality of language in use

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In the previous chapter, we looked at how languages differ among themselves, and how these parametric differences were described throughout the course of generative linguistics. The new, minimalist view of parameters works together with the specific language architecture and the view of multiple spell-out at phases smaller than a sentence. We considered the possibility that language variation resides in the lexicon, through the grammatical features reflected in various lexical items and in various combinations, but also at the interface with sound (PF) and with meaning (LF). In this chapter, the language architecture and the grammar continue to be very important; however, we consider their relationship with the processing system. We will be interested in the psychological reality of language during the encoding and decoding of the linguistic message. In current generative theory, linguistic structure, acquisition, and use are intrinsically associated (Truscott and Sharwood Smith 2004).

3.1 What happens when we hear a sentence?

What happens when we hear a sentence? We listen out for the sounds and, as they come in, we store them in short-term memory until we have enough to make a word. Then the word is passed on to the mental lexicon, the word storage space where its meaning, morphology, and syntax are activated, and the sounds making up the next word start coming through. Again, the results of lexical processing are retained in short-term memory. We start combining the words into a syntactic structure as soon as we can, and we continue this process (called *parsing*) until all the words have been accommodated in the structure. We clear up any uncertainties about who did what to whom using knowledge of the world and discourse context, and we finally reach the full sentence meaning. Is language comprehension really that simple? Not at all, but we will start this section with a breakdown of the processes in order to understand what is involved in decoding a linguistic signal. The short answer: Quite a lot. We will briefly discuss phonological, lexical, morphological, and syntactic processing in this chapter in order to delineate the main ingredients of linguistic processing.

3.2 Phonological perception and lexical recognition

Speech, not writing, is the primary medium of language. Many languages do not have writing systems, many others only developed such systems in the last century or two. Reading is a relatively new capacity from an evolutionary standpoint. Children are quite good at their native language before they learn how to read, and arguably, writing does not aid their language acquisition.¹ Having established that spoken and heard language is the primary modality for language acquisition as well as for most communication, we start our discussion of processing from the acoustic signal. We should distinguish between phonological perception and word recognition, the process that happens when a proficient native speaker hears a word

¹ Later on in life, after they learn to read and write, children definitely expand vocabulary through those means as well. They are also exposed to structures that are more typical in written language (such as the passive) but that they also learn to incorporate in speech.

that is part of a sentence. We will leave the issue of how children break into the acoustic signal of their native language for Chapter 4.

Adult speakers of English have been estimated to know between 50,000 and 100,000 words. They are stored in something that linguists call the *mental lexicon*, an organized system of all those words that we know. When we hear the acoustic signal of a word, we start a search of our mental lexicon. This process does not resemble the search for a word in a dictionary. The frequency of individual words and even word parts (morphemes) is crucial for our vocabulary knowledge. That is because vocabulary knowledge depends on experience. You are quite confident that you know common words such as *coffee* and *problem* while you might not be so certain about words like *commensurate* and *berm*. It is possible that you know you have heard a word, but you are unable to give a definition or paraphrase its meaning.

But what happens when we hear a word that is part of a sentence? Well, that depends on whether we speak the language of that sentence or not. Imagine that you are played a sentence in a language such as Kikuyu. You will not know where one word stops and the next begins. What we do in a language we speak is *segment* the speech into words. Hearing speech is not at all like seeing the words written on the page, because they are already segmented in the written text. As an exercise, take a look at the waveform of a sentence given in Figure 3.1.

Try to figure out where the word boundaries are. In a normal speech situation, it is not the case that we hear silent spaces between words and, as a reflection of that, we do not see gaps in the waveform at the end of words.

One could thus think of the speech signal as a text without spaces.

Another pervasive process in speech is *coarticulation*. The individual sounds change some of their qualities to blend together, to become more like the surrounding sounds, and that is true across word boundaries. Let us take an example of coarticulation with the pronunciation of the word *happy* whose phonetic representation is /hæpi/. Before you say anything, you will have moved your tongue into position for /æ/. Then, while you are saying /h/, it will sound a bit like /æ/. While you are saying /æ/, you will also be closing your lips for /p/. While your lips are together for /p/, you will be moving your tongue from where you had it for /æ/ to where you need it for /i/. Finally, while you are saying /i/, you will be opening your lips after /p/. The whole word will usually be uttered in less than half a second. To take another

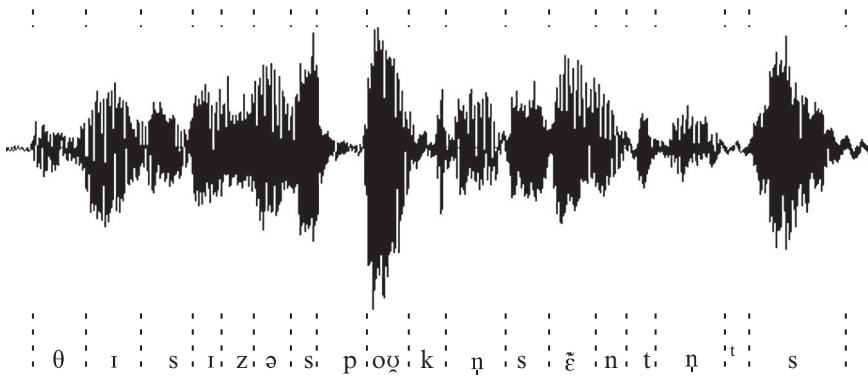


Figure 3.1 Spectrogram of the sentence *This is a spoken sentence*

example, in French, a word-final voiceless consonant will often become voiced if followed by a voiced segment. So the word *avec* ‘with’ on its own is pronounced /avek/, but when it is followed by a word beginning with a voiced consonant such as /v/ in *vous* ‘you’ /vu/, we usually hear /aveg/. So the phrase *avec vous* ‘with you’ is often pronounced /aveg vu/. In short, speech is a continuous, connected stream of sounds, and sounds have a strong effect on other sounds that are adjacent or close to them.

In order to understand speech, the listener has to break down the continuous speech signal into a discrete sequence of words. Even before the listener gets to the lexical information, there are pre-lexical characteristics of the speech stream that help to achieve segmentation. Chief among pre-lexical cues is the rhythm of speech, the repetition of sequences of stressed and unstressed syllables that give the speech signal its specific rhythmic sound, not unlike music. In tone languages such as Mandarin and Thai, tone would be another pre-lexical cue for segmentation.

However, speech rhythm can only take the listener so far in the breaking of the stream of sounds. Adult listeners get crucial help in this process from their native vocabulary. The speech segmentation process goes hand in hand with retrieving word forms stored in our mental lexicon. Indeed, models of spoken word recognition have tried to explain how we find matches between sequences of sounds in the speech signal and word forms in the lexicon (e.g., Marslen-Wilson and Welsh 1978, McClelland and Elman 1986, Norris 1994). In these models, parts of the speech signal are matched with words that we have stored in our mental lexicon, until every part of the speech signal corresponds to a meaningful word.

The simplest version of lexical recognition starts with mapping the incoming signal onto the phonological representation of existing words in the mental lexicon. The first psycholinguistic model to deal specifically with the recognition of speech is the Cohort model of Marslen-Wilson (1987). This model proposes that when the first 150–200 milliseconds of the word is encountered, or roughly the first two sounds, a *cohort* of other words matching the word up to that moment is activated. For example, if encountering the beginning of the word *February*, /fɛ/, all words beginning with /fɛ/ are activated such as *fence* and *phenomenon*. Note that spelling doesn't matter. When the subsequent sound comes in as /b/, words that do not begin with /fɛb/ drop out of the cohort. This process is repeated until (optimally) the cohort is reduced to one member. *February*, for instance, can be recognized by the third segment, because no other English word begins with /fɛb/. This moment is known as the uniqueness point. The recognition point in this model can come well before the actual end of the word. As more input is heard, some words in the cohort become incompatible with the input sound and are filtered out. This would be phonetic-based filtering. But filtering can also be contextual: words that are incompatible with the syntactic or semantic context are also filtered out.

This latter claim suggests that not only the phonological representation is activated at the very beginning. Marslen-Wilson (1987) demonstrated this with an experimental task called (cross-modal) priming. When experiment participants are asked to recognize a *target* word written on a screen as a word or a non-word, the task involves lexical decision. However, when another word is seen for a fraction of a second before the target word, this other word is known as a *prime* and it influences the speed of recognition of the target word. If the prime is heard and the target is written, priming is called cross-modal because it spans two modalities. Marslen-Wilson found that semantic similarity between prime–target pairs such as *confess* and *sin* leads to faster recognition times of the target, as compared to a non-related prime such as *tennis* and *sin*. The researcher wanted to find out whether cohort meanings would be activated before the uniqueness point. He primed both *sin* and *wedding* with the word fragment *confe*, which could be the beginning of *confess* as well as *confetti*. He found facilitation effects in both cases, which suggests that when we hear a word, we do not only activate the meaning of that word, but also, very briefly, the meanings of its competitors in the cohort. More recently, Gaskell and Marslen-Wilson (2002) argued

that the meaning activation is limited to just one or two meanings compatible with the speech input.

In the Cohort model, words are either a part of the word-initial cohort of activation, or they are not. However, other models of speech recognition, such as the TRACE model (McClelland and Elman 1986), introduced a refinement of the process. If a number of words are compatible with the initial portion of the word, but some are better candidates than others due to consistency with the input, then those better candidates will be more highly activated than the weaker candidates. Thus activation levels are on a continuous scale.

Mental lexicon researchers now agree on at least three aspects of spoken word recognition. First, there is consensus on the fact that multiple word candidates are activated in parallel as a word is being heard, constituting the word-initial cohort. Secondly, activation of word candidates varies with the degree of matching between the speech signal and stored lexical representations. Finally, all activated candidate words compete for recognition (Weber and Scharenborg 2012). Researchers also agree that word knowledge is the foundation on which knowledge of language rests.

Teaching relevance

Why is this important for teachers of language to know? It is important to appreciate how complicated the process of word recognition is in adult native speakers of a language, because one can then appreciate how much more complicated the process of word recognition is in bilinguals. Bilinguals have potentially double (although rarely so) the number of vocabulary items in their mental lexicon, some from their native and some from their second language. Imagine the mental lexicon of multilinguals. As a result of larger mental lexicons, the word recognition process in multilinguals takes longer. It makes sense that the following structure-building processes are going to be slower, too, compared to monolinguals. Note, however, that slower mental lexicon access does not really mean imperfect, or faulty, access.

3.3 Morphology

In the previous section, we discussed how words are picked out from the speech stream and accessed in the mental lexicon. In this section, we will look at another aspect of this same process: whether whole words or

morphemes are being accessed, and what the factors affecting this process may be. It is often taken for granted that we access the words of the incoming sentence, but actually there is compelling evidence that this is not always the case. Before we can explain why, we have to go over some terms and the related concepts.

As you know, words can be divided into *morphemes*, the smallest meaningful units of language, and these units follow certain (morphological) rules when they combine to make up words. For our purposes here, it is essential to distinguish functional morphemes from lexical morphemes (see also Exercise 2.3 in the previous chapter). In the word *book-s*, *book* is a lexical morpheme (it can also be a word on its own, so it is a *free* morpheme) while *-s* marks the plural. The latter cannot exist on its own so it is classified as a *bound* morpheme. Morphemes that add grammatical meaning (plural, tense, agreement, etc.) to the lexical items are called *inflectional*. Note that they cannot change the word category (N, V, etc.) of the word they attach to. If *book* is a noun, then its plural is still a noun, as in *He reads **book-s** all the time*. If *books* is a verb as in *She **book-s** her flights well in advance*, then the *-s* marks that the present tense verb agrees with a 3rd person subject (not plural). What is known as *functional* morphology includes inflectional bound morphemes as well as free functional words such as *the, a, of, and*, etc., that is, all morphemes with some grammatical function. All functional morphemes are non-category changing.

Derivational morphemes, on the other hand, expand the lexical meaning of the original lexical morpheme or word, and can change the word category. In the word *boy-ish-ness*, the adjectival suffix *-ish* attaches to the noun *boy*, while the nominal suffix *-ness* attaches to the new word *boyish*. The structure of the word is as shown in (1). In English, the category of the whole word is the same as the category of the rightmost morpheme, in this case, *-ness*. There are no exceptions to this rule of word formation.

- (1) [[[boy_N] + -ish_A] + -ness_N]

Before you go on, think (on your own or with classmates) about the morphological structure of the following words: *desirability*, *ungentlemanliness*, *cooperatively*, and *examinations*.

Do these linguistic descriptions hold any relevance to how the mental lexicon is organized? A full-listing approach would have it that all complex words are stored whole so that the polymorphemic word *disenchantment* is accessed as a whole unit, the same way as the monomorphemic word *cat*. (How many morphemes in *disenchantment*, by the way?) Almost no one believes this is the case, however. The opposite proposal would be the affix-stripping, or a fully compositional approach (Taft and Forster 1975), according to which all polymorphemic words are composed as they are being accessed, within the mental lexicon. As usual, the truth lies in between these two extreme proposals. It turns out that the most important factor affecting decomposition in the lexicon is a word's frequency: the more frequent a word, the higher the chance that it will be stored and accessed as a whole unit.

Decomposability is another factor, though. We will illustrate this with one of the longest and most important debates in contemporary cognitive science: the Great Past Tense Debate. As is well known, English has regular past tense forms such as *walk-ed* but also irregular ones such as *ate*. The latter is morpho phonologically opaque, meaning that it is not clear where the verb form stops and the past tense morpheme begins. In fact, it is quite clear that most irregular past tense forms are *not* decomposable into a lexical and an inflectional morpheme, although the two meanings are present in the form: *ate* = *eat* + [past]. These features of English past tense inflection allow researchers to contrast the whole-word and the compositional approaches to lexical representation in a direct way. The debate has been waging since the mid-1980s and is also known as the Words and Rules Debate. Its significance is enhanced by the fact that it has become a proxy for comparing and contrasting two fundamentally opposed approaches to linguistic knowledge. The fundamental choice is whether rule-like behavior should be explained in terms of symbolic computation describable with a rule such as *verb* + *-ed* = *past tense verb* (Pinker 1999, Pinker and Ullman 2002), or in terms of connectionist learning systems, containing neither rules nor symbols but only levels of activation based on previous experience (McClelland and Rumelhart 1986, McClelland and Patterson 2002).

It is important to mention that it is not irregular verb forms that are under debate: both sides of the debate agree that irregulars are acquired and processed by a connectionist (usage-based) learning mechanism. The disagreement comes in the treatment of regular verbs. The connectionist approach holds that both regular and irregular verb forms are controlled

by a single, integrated mechanism of word learning and access. This mechanism depends on phonological and semantic constraints operating together. The approach rejects the analysis that stem and inflectional morphemes are separable, and argues instead that inflected forms are learned and represented as whole forms. The more a verbal form (whether regular or irregular) is encountered and used, the stronger its associations within the neural network.²

On the other side of the debate, psycholinguists such as Pinker, Ullman, Marcus, Clahsen and colleagues argued vigorously in several papers over the years (e.g. Pinker and Prince 1988, Pinker 1999, Pinker and Ullman 2002), that regular past tense verb forms are decomposable, and that every time they are accessed, a symbolic rule operates in the lexicon to combine them into the needed form. In the mid 1990s, the nature of the evidence that the two sides were bringing to bear on the debate changed from studying behavioral observations and measures to looking at the underlying neural systems supporting the two types of forms. Marslen-Wilson (2007) reviews evidence from several studies investigating the behavior of people with brain lesions. These studies, according to him, point to the conclusion that decomposition processes are obligatory in the brain. They map the semantic and the phonological information of the verbal forms to brain regions that are separable although related. The on-line interpretation process integrates the use of these different brain regions. This conclusion can be extended (with caution) to all types of inflectional morphology in the mental lexicon.

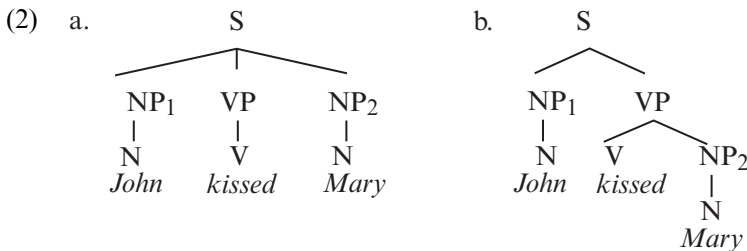
3.4 The syntactic parser

Morphemes and words are just the ingredients of sentences and discourse. In order to understand linguistic processing, we now look at the processing of syntax. Syntactic processing is executed by the *syntactic parser*. Parsing involves the rapid and automatic assignment of grammatical structure to a sentence encountered in speech (Pritchett 1992: 1). Without assigning structure to the incoming signal, comprehenders cannot understand who does what in sentences such as *John kissed Mary*. Interpretation depends, among other things, on *thematic roles* such as Agent and Patient (also known as

² If you want to learn more about connectionist neural networks and how they work, refer to this source: <http://plato.stanford.edu/entries/connectionism/>.

Theme). When we hear a verb, its objects, and its subject, we attribute such thematic roles (a.k.a. theta roles) to the noun phrases surrounding the verb, depending on its meaning. For example, John is the Agent of the Mary-kissing event, and Mary is the Patient (the event was initiated by John and may have been without her active participation). Other such theta roles are Experiencer (*John dislikes his new boss*), Recipient (*John gave Mary a check*), Goal (*Mary walked to the farmers' market*), and others. We normally interpret theta roles without much confusion, because they constitute lexical information encoded in the verb.

Now look at the two structures in (2a) and (2b) representing *John kissed Mary*. The labels of nodes we will use in this chapter conform better to psycholinguistic practice, and are somewhat simplified. S stands for Sentence, NP for noun phrase, and VP for verb phrase.



Note that the structure in (2a) presents the two NPs as equal partners, or sister nodes, to the VP. That structure predicts that the subject and object would behave in a similar fashion in various sentences. On the other hand, the structure in (2b) entails that the verb and the object come together, or merge first, and then as a second step, they merge with the subject. The relationship between the verb and the object is somehow closer than the relationship between the verb and the subject. This central syntactic fact is supported by a range of well-studied subject–object asymmetries, and we will glance at just two of them.

Marantz (1984) observed that while English has many idioms and specialized meanings for verbs in which the internal argument, the object, is the fixed part of the idiom and the external argument, the subject, is free, the reverse situation is considerably rarer. To put it in other words, the role played by the subject argument often depends on the filler of the object position, but not vice versa. Take Kratzer's (1996: 114) examples:

- (3) a. kill a cockroach
b. kill a conversation
c. kill an evening watching TV
d. kill a bottle (i.e., empty it)
e. kill an audience (i.e., wow them).

The killer of a bottle is a drinker, while the killer of an audience is a performer. As the examples above illustrate, the meaning of the verb within the verb phrase changes dramatically depending on the object, while this rarely happens when we vary the subjects. This fact supports structure (2b) as the correct analysis.

Secondly, we can have a reflexive pronoun in object position but not in subject position, and this is true universally for all languages. The reflexive *himself* is not self-sufficient in meaning: it takes its reference from its antecedent, in this case the NP₁ *John*. The antecedent has to *c-command* the reflexive, otherwise the sentence would sound awful as in (4b). C-command is a basic relationship between nodes in a tree structure. A node, say NP₁ c-commands another node NP₂ if NP₁ does not dominate NP₂, NP₂ does not dominate NP₁, and the first branching node dominating NP₁ also dominates NP₂. See for yourself how these relations play out above in trees (2a) and (2b). C-command has not only syntactic consequences, but interpretive ones, too.

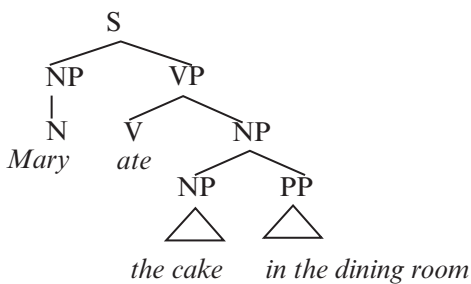
- (4) a. John likes himself.
b. *Himself likes John.

If our structure of a simple sentence were as in (2a) with NP₁ and NP₂ being on an equal footing, we would not be able to explain these facts, and especially the closer connection between the verb and its object. Remember, structure predicts syntactic behavior and interpretation. This is a potent fact.

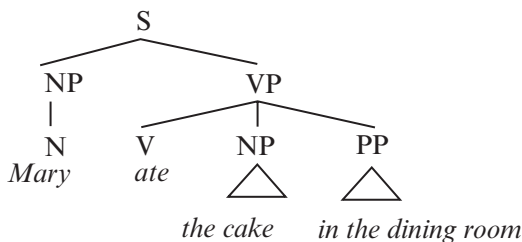
When we have an ambiguous sentence, a sentence with two interpretations, unless the ambiguity is lexical,³ it is usually diagrammed as two distinct phrase structure trees. Take for example the sentence in (5).

³ An example of a lexical ambiguity are the two meanings of the word *port*: “haven” and “wine.” These two meanings are equally possible in the sentence *They passed the port at midnight*. What are the two meanings: paraphrase them. Of course, in normal communication, the context usually disambiguates.

- (5) Mary ate the cake in the dining room.
- Interpretation 1: Mary consumed the cake that was in the dining room (but she may have left untouched the cake that was in the kitchen).
 - Interpretation 2: Mary engaged in a cake-eating event in the dining room. It was in the dining room where that event happened.
- (6) Structure for Interpretation 1. (A triangle represents more detailed structure inside the node, but it is not shown because it is not relevant for the particular point we want to make.)



- (7) Structure for Interpretation 2



In (6), the PP *in the dining room* modifies only the NP *the cake*, pointing to where that sample of baked goods is to be found. In (7), however, the same PP modifies the verb and the NP object together, signaling that the eating event happened in the dining room. See how structure determines interpretation?

The representations in (6) and (7) are linguistic representations. But are they used by the parser? Psycholinguists argue about the relationship between real-time computation (parsing) and the underlying grammar (or grammatical representations). While the issue has not been solved

definitively and to everyone's satisfaction, the gamut of views ranges from a minimal relationship, or even a complete separation, to a functional dependency. Thus, one view is that the grammar just describes grammatical and ungrammatical sentences (through its rules and constraints) but is not directly implicated in real-time language processes (Sag and Wasow 2011). On the other hand, another view (Pritchett 1988, 1992, Phillips 2013) is that there is no need for the grammar and the parser to be separate, that they are aligned and the grammar can be implemented in on-line comprehension and production, with the addition of other psychological mechanisms. Juffs and Rodriguez (2015: 15) provide a useful analogy for the relationship of the grammar and the parser as the internal combustion engine at rest versus running. Do you see a parallel with Mark Baker's bread versus bread recipe metaphor?

[T]he grammar is the engine at rest, not driving the vehicle, but with the potential to do so. Parsing is the engine in motion, subject to stresses and possible breakdowns allowable by the system, and driving production or comprehension in real time. Just as an engine's function can be affected by fuel, availability of oxygen, oil level and road conditions, the operation of the grammar during processing may be affected by the quality of input, memory limitations, and interference from outside influences not related to the architecture of the grammar itself.

Current evidence favors the view that human grammatical abilities are best understood as a single structure-building system that assembles syntactic structures in a roughly left-to-right order and that the internal grammar is a key component of the parsing and production mechanisms. We shall now look at some concrete proposals. Consider again the ambiguous string of words in (5) and its two interpretations in (6) and (7). The sequence of words can be structured, or organized, in more than one way, consistent with the grammar rules of the language. Are ambiguous sentences harder to understand, compared to unambiguous sentences of similar word and structure complexity? If the answer is yes, then this fact would support the hypothesis that structure is indeed quickly built in parsing sentences. It could also be used to argue for an actual separation between the parser and the grammar, with the grammar proposing structures and the parser taking these structures for a test drive to choose the better option.

The short answer is that ambiguity does lead to longer reading times, and lower comprehension accuracy, although the preceding context can substantially modify these effects. In order to understand how listeners and readers attribute structure to sentences, linguists study the parsing of

temporarily ambiguous sentences (also known as *garden path* sentences) such as (8).

- (8) The tomcat curled up on the cushion seemed friendly.

Temporarily ambiguous sentences contain a sequence of words that can be plausibly arranged in one structure, but half-way through the computation, the incoming words suggest that the first analysis was wrong and has to be changed. The comprehender ultimately arrives at just one global sentence interpretation, but was “led down the garden path” for a short time during the parsing. Imagine that you are reading on a screen the words or phrases of sentence (8), appearing one after the other, in the exact spot where they would be if the whole sentence were put up on the screen. This is known as the self-paced moving window presentation paradigm, because readers have to press a button every time they are ready for the next word to appear on the screen. It is assumed that the time between the appearance of a word and the button press for advancement is a function of how easy or difficult it is to accommodate the last word within the structure already built in the mind by that moment. Reaction times are calculated for each word or phrase slot.

- (9) Self-paced moving window presentation

- a. The tomcat _____
- b. _____ curled up _____
- c. _____ on the cushion _____
- d. _____ seemed _____
- e. _____ friendly.

Imagine that the sentence starts with step a), then step b). By the end of step c) *on the cushion*, the parser would be ready to fold up with the structure since this is a plausible sentence: *The tomcat curled up on the cushion*, with a subject NP, an intransitive verb and a location prepositional phrase (PP) analysis. When the next verb appears in step d), however, it becomes immediately clear that *seemed* is the main verb of the sentence and the grammatical function of *curled up* has to be reconsidered. The parser has to perform the additional step of reanalysis of the preceding structure into a reduced relative clause: *The tomcat (that was) curled up on the cushion*, where the words in parentheses are optionally omitted. That additional step

takes longer to accommodate. In general, the longer it takes people to understand part of a sentence, the greater the processing load it imposes. When the final segment *friendly* appears, it just confirms the second analysis.

Before going further, try to analyze the following sentence (from Frazier and Rayner 1982) in such an incremental way:

(10) While Susan was dressing the baby played on the floor.

What would be the first analysis, which segment would set off the reanalysis, and what would be the second analysis? Think about the properties of the verb *dress*. What property of this verb makes the temporal ambiguity possible?

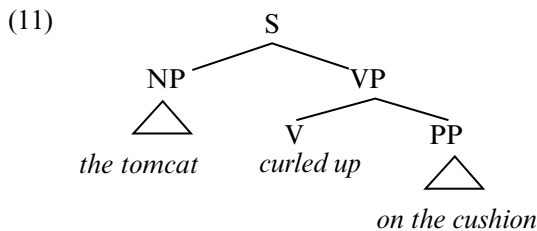
The view that the language processor operates incrementally has enjoyed a lot of experimental support: it rapidly constructs a syntactic analysis for a sentence fragment, starting from left to right, merging syntactic constituents to establish a sentence node and then to expand it to the right. The processor also assigns each constituent a semantic interpretation, and relates this interpretation to world knowledge. In the next section, we will go over the classic models and debates of syntactic processing.

3.5 Models of syntactic processing

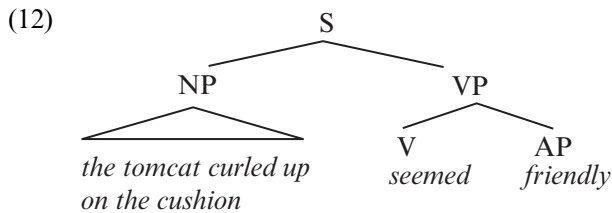
There is considerable controversy about when language users employ different sources of information during sentence processing. There is a major divide in the psycholinguistic literature depending on the following fundamental question: whether hearers/readers immediately use all relevant sources of information, or whether some sources of information are delayed relative to others? Based on the answers to these questions, sentence-processing theories can be divided into *interactive* accounts, in which all relevant information can be used immediately, and *modular* accounts, in which some information can be used immediately but some information cannot, and parsing has a somewhat cyclical nature.

We shall start by exemplifying the latter, among which the Garden Path model (e.g., Frazier 1987, Rayner et al. 1983) is the classic and most

influential account. What does it mean that readers or listeners slow down their processing at step d) in example (9) above? It means that they have been making decisions about how to organize words into phrases and clauses before they have received sufficient information to be certain about the decisions they are making. In the moving window presentation, readers see that there are two more segments coming up, but that doesn't stop them from positing an analysis of a complete clause with the words that have appeared until that moment. Readers as well as listeners follow the *immediacy principle*: build as much structure as you can, as soon as possible (Just and Carpenter 1980). In the case of sentence (9), by the time they have heard step c) they build structure as in (11):



Only after they encounter the verb *seemed* do they go back and reconsider that analysis to come up with the correct one in (12).



The idea that listeners and readers build and maximize structure immediately, even if it is the wrong structure, then get stranded along the garden path, after which they go back and reconsider their initial analysis at considerable processing cost, is at the foundation of Lyn Frazier's Garden Path model (Frazier 1979, 1987, Frazier and Rayner 1982). It is considered a *two-stage processing model* because she proposes that syntactic processing goes through an initial stage, at which the incoming words are analyzed to see what category they belong to: A, N, V, P, etc. No other information is

available at that first stage, the parser only cares about categorical information and phrase structure rules. At the second stage of sentence interpretation, meaning is computed by applying semantic rules to the syntactic structure already built. That is why semantic information, frequency, and information from the context can only be taken into consideration at this second stage of processing. If we consider the debate mentioned in the beginning of this section, namely, at what stage of sentence processing people use different sources of information, the Garden Path model will be classified as a modular account. Modular models assume that the mind consists of modules that perform very specialized processes, and these processes are informationally encapsulated (closed off) in the sense that they use only information represented within that particular module (Fodor 1983).

The classic version of the model proposes two parsing heuristics, *late closure* and *minimal attachment*, that can be seen as embodying a drive for processing efficiency. Late closure postulates: Do not build more structure than strictly necessary, if possible continue to work on the structure that you are currently building.

Going back to our examples, try to figure out how late closure explains the wrong initial analysis of sentence (9) as in (11).

Minimal attachment urges the parser to create structure with as few nodes as possible. In later work, these heuristics were augmented with the *main assertion preference*, which proposes that if the parser has any choice, it builds a structure where the new elements are related to the main assertion of the clause. In other words, they are arguments, not adjuncts.

Recent parsing findings suggest that the parser applies the heuristics in a flexible way. What heuristics are going to be applied in the sentence analysis will depend on the properties of the sentence, and most importantly, on the selectional requirements of the verb. As Traxler (2012: 151) points out, one of the main advantages of the Garden Path model is that it makes concrete predictions on where readers and listeners will experience processing difficulty. That means that the model is falsifiable, which is a fundamental requirement of all scientific proposals (think of the scientific method discussion in Chapter 1).

Alternative models of sentence parsing are *constraint-based*, or *interactive* parsing models (e.g., MacDonald, Pearlmutter, and Seidenberg 1994, Spivey-Knowlton and Sedivy 1995, Trueswell, Tanenhaus, and Garnsey 1994). Rather than building one structure at a time, constraint-based parsers are capable of pursuing multiple analyses of a clause at the same time. This is because the models adopt a parallel distributed processing architecture, similar to the TRACE model we discussed of lexical processing (see Section 3.2). This type of parser represents phrases and clauses not as grammatical structures as in (11) and (12) above, but as patterns of activation spreading over interconnected processing units. The units make up a neural network resembling the functioning of networks of neurons in the brain. While partial information can lead to partial activation of multiple mental representations, experience, frequency of association, and context decide which of the partially activated structures is the most likely analysis.

To come back to the example in (9), the constraints allow the parser to start building several structural hypotheses, using information from the lexical items, as well as syntactic and semantic knowledge, all at the same time. Because of this simultaneous employment of all possible resources, constraint-based models are known also as *one-stage models*. Of the multiple analyses being entertained, the more plausible ones are awarded with more activation while the less plausible ones lose out until they fade out completely. The activation levels presumably change with every incoming word. When two analyses remain highly plausible, hence activated, it takes a long time (relatively speaking) before the correct analysis wins the competition and the incorrect (but initially highly activated) analysis is inhibited. Thus competition results in processing difficulty and delay. Note that we cannot speak of reanalysis with this type of model, since both analyses are activated from the onset of the ambiguity. An attractive feature of the interactive processor allows it to immediately draw upon all possible sources of information during sentence processing, including semantics, discourse context, and information about the frequency of syntactic structures.

Most constraint-based models are also *lexicalist* in the sense that they assume that syntactic information is associated with words. As mentioned above, it is generally well established that verbs contain information about the arguments that they select. In addition, all verbal lexical entries are assumed to encode the frequency with which they occur in particular argument frames. For example, the verb *see* appears 9 times out of 10 with a direct object argument (e.g., *John saw Mary*). In 1 out of 10 cases,

the verb is used intransitively (e.g., *I see*). This type of information is used during syntactic ambiguity resolution. Hence, many constraint-based models assume that there is a tight correspondence between sentence comprehension and production preferences: Structures that are produced frequently should be easier to process than structures that are infrequent. In a nutshell, these models highlight the role of experience and not so much linguistic creativity.

Next, let us delve deeper into the issue of whether and when semantic information is taken into account to constrain syntactic analysis, in a sense, to guide sentence processing. The following discussion is an illustration of how linguistic models make predictions about behavior, and how existing accounts are amended in order to accommodate new data. Semantic information can provide strongly constraining information over syntactic and lexical analyses in the cases of ambiguity.

(13) They passed the port at midnight.

In the ambiguous sentence in (13), if the previous discourse context is about a party with various drinks being served, then the ‘fortified wine’ meaning of *port* will be much more activated than the ‘haven’ meaning. In cases of temporary syntactic ambiguity such as reduced relative clauses, the constraint-based models predict that semantic information such as animacy has an immediate effect on processing and even allows the comprehender to anticipate upcoming syntactic structure. The Garden Path model, on the other hand, would not predict such immediate effect, since according to this model the parser ignores semantic information as it is making its initial syntactic decisions. A number of studies have examined the effect of animacy and selection frequency in sentences such as the ones in (14), again involving a reduced relative clause:

- (14) a. The defendant examined by the lawyer turned out to be unreliable.
b. The evidence examined by the lawyer turned out to be unreliable.

Sentences such as in (14a) are predicted to be hard by the Garden Path model because the first two words *the defendant examined* are likely to be immediately merged in an SVO structure “the defendant examined somebody” and not as the more complex reduced relative structure “the defendant who was examined.” Furthermore, the model does not predict

significant differences between sentences (14a) and (14b), as they have similar syntactic structure. The constraint-based model, on the other hand, predicts that the animacy of the subject in (14a) will create problems for comprehenders, because people and other animate things are more likely to initiate actions, move around and generally be movers and shakers, that is, thematic Agents. Therefore, a construal such as “the defendant examined somebody” is highly activated. The reduced relative construal, “the defendant who was examined,” is also activated. When the parser reaches *by the lawyer*, the Agent assumption has to be abandoned. The defendant is not the initiator of the examining in this sentence, but its recipient, hence Patient or Theme; the second construal wins over the first one. However, the case in (14b) is much simpler: *evidence* is an inanimate object and thus not a likely Agent. On encountering a sentence such as (14b), a comprehender should not activate an Agent-plus-active-verb analysis very highly.

Thus, we have two contradicting predictions for these two sentences by the two leading parsing models. The experimental evidence comes down on the side of the fast contribution of the semantics. In experiments comparing the parsing of sentences such as (14a) and (14b), Clifton et al. (2003) and Trueswell et al. (1994) found that sentences with an initial animate noun do indeed impose more processing difficulties than sentences with an inanimate noun. Such findings, then, represent a problem for the Garden Path model.

In later work, Frazier and Clifton (1996) refined the Garden Path model to include the idea that the parser may sometimes build parallel structure simultaneously and that context may sometimes influence the parse very fast. One case in which this happens is in a sentence such as (14b), where the inanimate noun helps the parser reject that wrong structure very quickly, so that it can start building the correct structure almost immediately. Recently, alternative models have appeared that refine our understanding of the parser to include attractive features of both Garden Path and constraint-based models.⁴

An altogether different approach to native language processing is seen in the model proposed by Edward Gibson. Gibson’s (1998, 2000) Dependency Locality Theory focuses not so much on sentence ambiguity but on structural complexity. At the bottom of the theory is the well-established finding

⁴ E.g., the unrestricted race model, Van Gompel, Pickering, and Traxler (2000), Van Gompel, Pickering, Pearson, and Liversedge (2005).

that object relative clauses are more difficult to process in English than subject relative clauses, see examples in (15a,b).

- (15) a. The reporter [who ____ attacked the senator] admitted the error.
(Subject Relative)
b. The reporter [who the senator attacked ____] admitted the error.
(Object Relative)

Square brackets mark the relative clauses, while the underline marks the original position of the *wh*-word. In (15a) the reporter is the one who attacked someone, while in (15b) he was attacked. Since there is no ambiguity at any time, the robust finding of differential processing difficulty is hard to explain with parallel activation levels or structure reanalysis. Gibson proposes that these two types of relative clauses are processed differently because they vary in processing complexity. Two factors contribute to processing complexity: storage cost and integration cost. In other words, it is costly to store words in memory while processing, and it is cheap to integrate them as soon as possible. The former cost is incurred when a syntactic dependency between two words is *predicted* by the parser, and that dependency crosses over intervening referents. For example, in (15b) the relative pronoun *who* is thematically related to the verb *attacked*, but there is another intervening argument between them: *the senator*. Note that this is not the case in (15a), at least on the surface. If such a dependency has to be maintained in short-term memory during processing, the one in (15a) is shorter and less costly. On the other hand, integration costs arise when a syntactic dependency is *established*. On this count, too, the dependency in (15b) is costlier, since *the senator* intervenes between *who* and *attacked*. Gibson's model has been very influential in accounting for native language processing.

3.6 Working memory

Gibson's Dependency Locality Theory discussed in the previous section has given you a taste of how important working memory is for sentence processing. "Working memory" refers to the temporal storage and manipulation of linguistic elements in the course of language processing, and is thus different from our long-term memory. As illustrated above, it involves both

storage and processing of information. Working memory has been operationalized by psycholinguists and cognitive scientists as two different constructs: phonological short-term memory (Baddeley 1999), and reading span memory (Daneman and Carpenter 1980). The phonological short-term memory is tested by asking research participants to memorize unrelated digits, real words, or nonce (made-up) words. In a reading span test, participants read aloud sentences and are required to remember the last word of each sentence, up to a set of six sentences. Thus, this working memory test claims to check both processing (the reading of sentences) as well as storage capacity (the memorization of words). Some indication that the two types of tests may access different memory systems is suggested by the fact that traditional span measures (using digits, words) do not decline with age and do not correlate with sentence comprehension impairment, whereas reading span does decline with age and correlates with sentence comprehension scores (Carpenter, Miyake, and Just 1994: 1078). While no consensus has been reached on whether the two memory tests access different memory capacities, most psycholinguistic research on native language processing uses the reading span measures, since this research often involves reading (Juffs and Rodríguez 2015: 45–46).

Much recent debate has centered around the question of whether the working memory resources employed during syntactic processing are different from the working memory resources used for other, more conscious verbal tasks. Just and Carpenter (1992) proposed the *shared resources hypothesis*, according to which all linguistic tasks draw on the same limited resource of working memory resources. The consequence is that when working memory is taxed or its capacity is exceeded, a slow-down in processing or excessive errors in judgments should result. A further consequence is that individual differences in working memory capacity among readers and listeners should lead to measurably different behavior.

In contrast, Caplan and Waters (1999) proposed the *dedicated resources hypothesis*: some working memory resources are dedicated to obligatory and automatic linguistic processes (such as parsing) while different resources are used for more strategic and controlled linguistic processes such as keeping words in memory. Thus, if this account is correct, one would not expect to find a correlation between accuracy on a processing task and a reading span measure. Experimental results to date are mixed, but on balance, there is more support for the dedicated resources account. We shall look at the results

of just one recent study, Sprouse, Wagers and Phillips (2012), on the connection of working memory and processing of islands.

Recall from Chapter 2 that “islands” are such linguistic structures from which a constituent word cannot be moved, or extracted. In examples (16a,b),⁵ the extraction of the *wh*-word from the object position of the lowest clause is acceptable. The underlying position of the moved element (or the gap, in psycholinguistic parlance) is marked by an underscore. Examples in (17) illustrate, however, that not all extractions are acceptable, so that there are various constraints on this movement. In other words, those embedded clauses in (17) are islands. For example, extraction is not allowed from subjects, from adjuncts, from complex NPs, and when the lower clause is topped off with the word *whether*.⁶

In order to understand these structures, try to amend the sentences in (17) to get acceptable structures. What did you change?

- (16) a. What does Susan think that John bought ____?
 b. What does Sarah believe that Susan thinks that John bought ____?
- (17) a. whether island
 *What do you wonder [whether John bought ____]?
 b. Complex NP island
 *What did you make [the claim that John bought ____]?
 c. Subject island
 *What do you think [the speech about ____] interrupted the TV show?
 d. Adjunct island
 *What do you worry [if John buys ____]?

It has recently been proposed (Kluender 2004, Hofmeister and Sag 2010) that there are no grammatical constraints on extraction, but instead long-distance dependencies are too complex to process, and their unacceptability is due to the breakdown of the parser while reading or hearing them. Such

⁵ The examples are from Sprouse et al. (2012: 82).

⁶ The embedded clause can be topped off with any other *wh*-word moved there within the clause, as illustrated in Chapter 2, example (12), **What did Jane wonder where Bill put?* The more general name for this type of island is a *wh*-island.

an idea entails that one will find a correlation between working memory capacity measures and accuracy on islands such as (17a,b,c,d). Sprouse et al. (2012) tested over three hundred native speakers of English on the four island-effect types exemplified in (17), using two different acceptability rating tasks and two different measures of working-memory capacity. The researchers found no evidence of a relationship between working-memory capacity and island effects. Their findings suggest that island effects are more likely to be due to grammatical constraints than to limited processing resources.

3.7 The psychological reality of language and the grammar

Over the last several decades, there has been an enormous increase in interest in the psychology and neural substrates of language processing. Accordingly, there has been a proliferation of studies using new experimental techniques such as functional magnetic resonance imaging (fMRI), event-related brain potentials (ERPs), and eye tracking. They have revealed a great deal about how and when different types of information are integrated during real-time comprehension by native speakers of a language. We will discuss these new techniques when we focus on L2 processing in Chapter 12. However, this is probably a good place to explain why language teachers and language learners should know about processing language.

As I mentioned earlier in this chapter, the debate on the relationship between the parser and the grammar is not yet completely resolved. However, thinking about the psychological reality of language use (what psycholinguists engage in) has already influenced evolving linguistic theory (what theoretical linguists do). To take some examples, the language architecture proposed by Jackendoff (2002) takes language use into account.⁷ Multiple spell-out, the idea that sentences are chopped into smaller chunks called phases and these chunks are sent to the two interfaces for interpretation right away, also reflects the computational needs of the processing system. Phases appear to offer computational advantages, enforcing locality of syntactic movement, thus reducing the information that must be held in “active memory” (Chomsky 2001, 2008). Finally, considerations of

⁷ As a result, it is taken as a basis for a number of SLA proposals, such as the Bottleneck Hypothesis and MOGUL (Sharwood Smith and Truscott 2014).

processing limitations and excessive loads are now very frequent in explaining why child or adult language learners are taxed and challenged by certain constructions and certain properties. We shall see such analyses later in this textbook. The point that I want to emphasize here is that grammar and acquisition are difficult to discuss for a long time these days without mentioning processing. Theoretical proposals about grammar and especially about language acquisition have to be psychologically real. If this textbook had been written ten or fifteen years ago, it would not have had two whole chapters on psycholinguistics. That is why language-teaching practitioners should keep abreast of developments in this fast-growing field.

Linguistic theory is essential in these developments. The new approaches and the new techniques are most effective in elucidating human language processing when they are based on well-understood linguistic distinctions such as the distinction between lexical and functional morphemes, between nouns and verbs, between main and embedded clauses, etc. Although there are still significant disconnects between linguistic theory and the neurobiology of language (Poeppel and Embick 2005), there are also promising areas of investigation where the neurosciences intersect with linguistic research more closely than before (Poeppel, Emmorey, Hickok, and Pylkkänen 2012). Although much of the integration between linguistics and neuroscience lies in the future, there is no turning the clock back.

3.8 Exercises

Exercise 3.1. Read the following sentences and discuss whether each one offers a lexical or a structural ambiguity. Some of the examples have more than two meanings, and some may combine lexical and structural ambiguity.

- (1) He gave her cat food.
- (2) I saw her duck.
- (3) They are hunting dogs.
- (4) Wrap poison bottles in sandpaper and fasten with scotch tape or a rubber band. If there are children in the house, lock them in a small metal box. (*Philadelphia Record*)
- (5) LOST Antique cameo ring, depicting Adam and Eve in Market Square Saturday night. (*Ad in Essex paper*)
- (6) He strips the bark and leaves.

- (7) “You know, somebody actually complimented me on my driving today. They left a little note on the windscreen; it said, ‘Parking Fine.’ So that was nice.” (Tim Vine)
- (8) “*Outside* of a dog, a book is a man’s best friend; inside it’s too hard to read.” (Groucho Marx)
- (9) The Rabbi married my sister.
- (10) She is looking for a match.
- (11) The fisherman went to the bank.
- (12) I know women more beautiful than Julia Roberts.

Exercise 3.2. Crash blossoms.⁸ The following are headlines from newspapers, which follow somewhat different conventions from normal sentences: they can be missing articles and other functional words for the sake of brevity. However, that allows for an unusually large amount of ambiguity. Discuss the following structurally ambiguous headlines, and more specifically, how lexical polysemy allows the ambiguity.

- (1) Kids Make Nutritious Snacks
- (2) Miners Refuse to Work After Death
- (3) Teacher Strikes Idle Kids
- (4) Red Tape Holds Up New Bridge
- (5) Local High School Dropouts Cut in Half
- (6) Stolen Painting Found by Tree
- (7) Include Your Children When Baking Cookies
- (8) Doctor Testifies in Horse Suit
- (9) American Ships Head to Libya
- (10) Enraged Cow Injures Farmer with Ax
- (11) Gadhafi Forces Barrel Into Main Rebel Base
- (12) Sting Proves Bing Copied Search Results
- (13) Squad Helps Dog Bite Victim

⁸ “Last August [2009], . . . in the Testy Copy Editors online discussion forum, . . . Mike O’Connell, an American editor based in Sapporo, Japan, spotted the headline ‘Violinist Linked to JAL Crash Blossoms’ and wondered, ‘What’s a crash blossom?’ (The article, from the newspaper *Japan Today*, described the successful musical career of Diana Yukawa, whose father died in a 1985 Japan Airlines plane crash.) Another participant in the forum, Dan Bloom, suggested that ‘crash blossoms’ could be used as a label for such infelicitous headlines that encourage alternate readings, and news of the neologism quickly spread.” (Ben Zimmer, “Crash Blossoms,” *The New York Times Magazine*, Jan. 31, 2010.)

- (14) Judge to Rule on Nude Beach
- (15) Police Discover Crack in Australia.
- (16) Satellite Tracks Cows From Outer Space

Exercise 3.3. Consider these garden path sentences.

- (1) Fat people eat accumulates.
- (2) The cotton clothing is usually made of grows in Mississippi.
- (3) Until the police arrest the drug dealers control the street.
- (4) The man who hunts ducks out on weekends.
- (5) When Fred eats food gets thrown.
- (6) Mary gave the child the dog bit a bandage.
- (7) The girl told the story cried.
- (8) I convinced her children are noisy.
- (9) Helen is expecting tomorrow to be a bad day.
- (10) The horse raced past the barn fell.
- (11) I know the words to that song about the queen don't rhyme.
- (12) She told me a little white lie will come back to haunt me.
- (13) The dog that I had really loved bones.
- (14) The man who whistles tunes pianos.
- (15) The old man the boat.
- (16) The tycoon sold the offshore oil tracts for a lot of money wanted to kill J.R.
- (17) The raft floated down the river sank.
- (18) We painted the wall with cracks.

Work on adding the disambiguating words in brackets, so that the sentences are no longer garden paths. The first example is done for you.

(The) fat (that) people eat accumulates. (So true!)

Some of the sentences in exercises 1–3 are borrowed from the following websites:

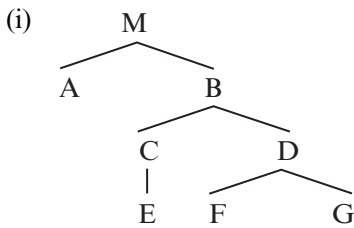
<http://www.byrdseed.com/ambiguous-sentences/>

http://www.fun-with-words.com/ambiguous_garden_path.html

<http://allthingslinguistic.com/post/36385656700/my-favourite-garden-path-sentences>

There are numerous blogs and websites devoted to fun with ambiguities and garden paths out there.

Exercise 3.4. C-command is a fundamental relationship between nodes on a syntactic tree, but it is also essential in language processing. Practice c-command. Look at the following tree and state the relationship between the nodes, answering the questions exhaustively. Refer to the definition of c-command above.



What does M c-command?

What does A c-command?

What does C c-command?

What does E c-command?

Does D c-command A?

Does B c-command A?

Does E c-command C?

Does E c-command D, F, and G?

Does F c-command G?

Does G c-command F?

Exercise 3.5. To illustrate how psycholinguists make predictions about various models, let's look at some predictions following the classic Garden Path model. Consider the sentences in (i) and (ii):

- (i) The burglar blew up the safe with the rusty lock.
- (ii) The burglar blew up the safe with the dynamite.

The sentence in (i) should be ambiguous. Look at the structures (6) and (7) in the chapter and say whether they can accommodate the ambiguity. Count the nodes of the structure it takes to represent the intended meaning.

Paraphrase each one, giving an explicit version of the meaning. Which structure, (6) or (7), is simpler, i.e. has fewer nodes? Which meaning would the minimal attachment principle predict that readers would assume? Will the parser then send an error message, and why?

Would sentence (ii) be easier to parse, and why?

It makes sense to assume that speakers would use a sentence like (i) with the meaning as in (6) only if there are two safes in the available context, and we have to distinguish one from the other, right? In this sense, if a test sentence such as (i) is introduced by context as below, that might help to disambiguate it (Altmann and Steedman, 1988):

A burglar broke into a bank carrying some dynamite. He planned to blow open a safe. Once inside, he saw that there was a safe with a new lock and a safe with an old lock.

In Altmann and Steedman's Experiment 1, sentences such as (i) were read **more quickly** preceded by such context. What does this suggest for the Garden Path model? Why? Does this finding support constraint-based models?

Answer: The parser does not automatically take the simpler analysis as in (7), the so-called VP-attachment analysis, if the context favors an NP-attachment analysis as in (6), even though the latter has more nodes. Hence, the minimal attachment does not obtain, because the context militates against it.

Exercise 3.6. It would go against constraint-based parsing models if the parser sometimes favors less likely structures over more likely ones. Consider the sentence in (i):

(i) The athlete realized her shoes somehow got left on the bus.

The athlete realized her shoes is not a plausible sentence, right? That's because the verb *realize* is most often followed by a complete clause, such as *She realized that she was hungry*. We can assume that this verb's preference to be followed by a clause would influence the parser right away and readers would not slow down at the noun *the shoes*. How would that follow from constraint-based models? In the eye-tracking experiment reported in Pickering, Traxler, and Crocker (2000), readers did slow down on *the shoes*. What does this finding suggest? How does it support the principle of minimal attachment?